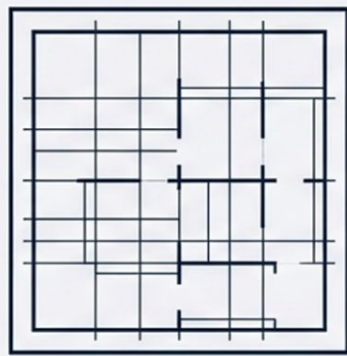


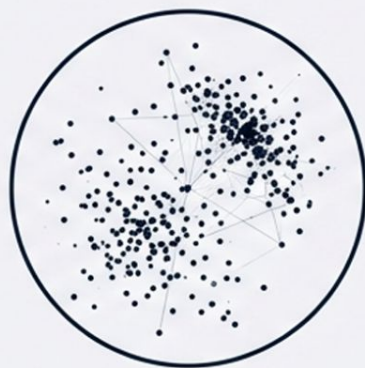
The Logic Shift: Rules, Learning, and Generation

AI has not just improved; it has fundamentally changed how it operates.



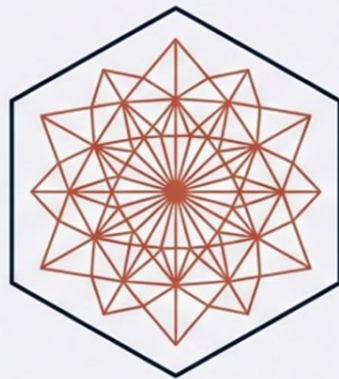
1. Rules

Systems followed strict, pre-programmed instructions.



2. Learning

Systems began analyzing data to find patterns.



3. Generation

Systems can now create new content based on those patterns.

The Foundations (1950s – 1980s)

1950s



The Turing Test

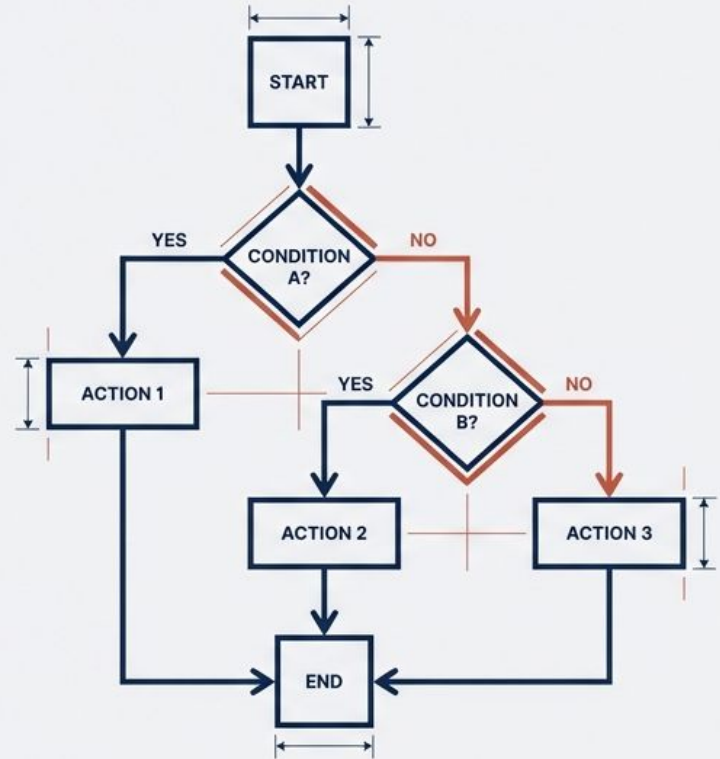
The theoretical birth of determining machine intelligence. Can a machine deceive a human?

1960s – 1980s



Rule-Based Systems

AI operated on strict “if-then” logic hardcoded by humans. These were **expert systems** designed for rigid decision trees.



The Data Era (1990s – 2000s)

1990s



Machine Learning Introduction

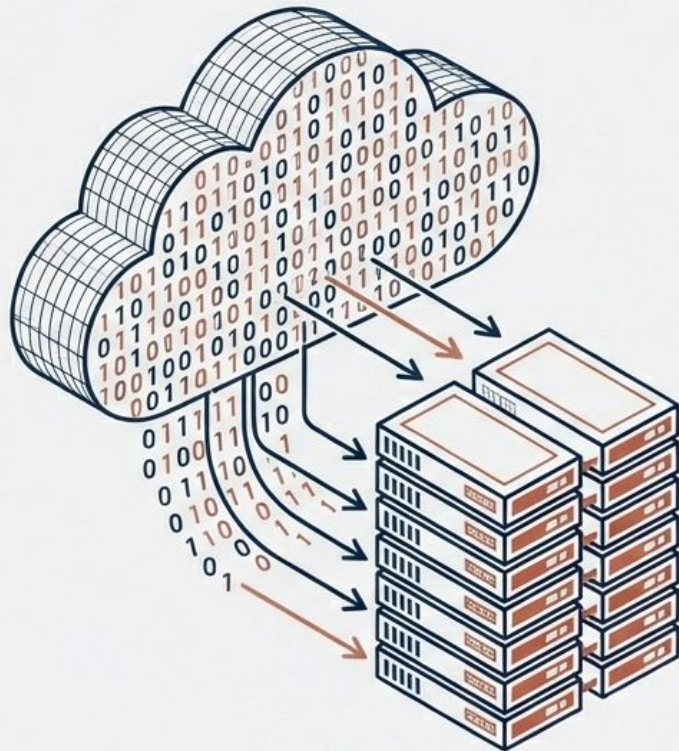
Computers begin moving beyond static rules to dynamic learning, finding their own patterns in input data.

2000s



The Big Data Explosion

The digitization of the world provides the massive fuel required for training learning algorithms.



The Deep Learning Revolution (2010s – Today)

2010s

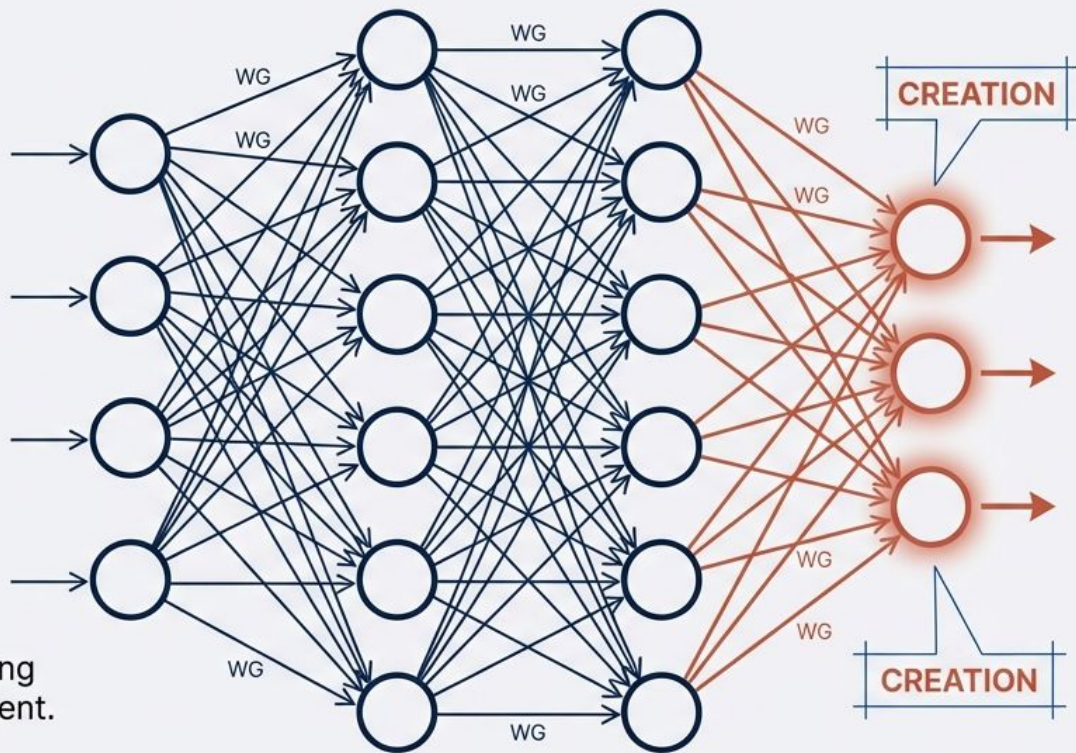
Deep Learning

Neural networks with many layers enable complex pattern recognition (vision, speech).

Today

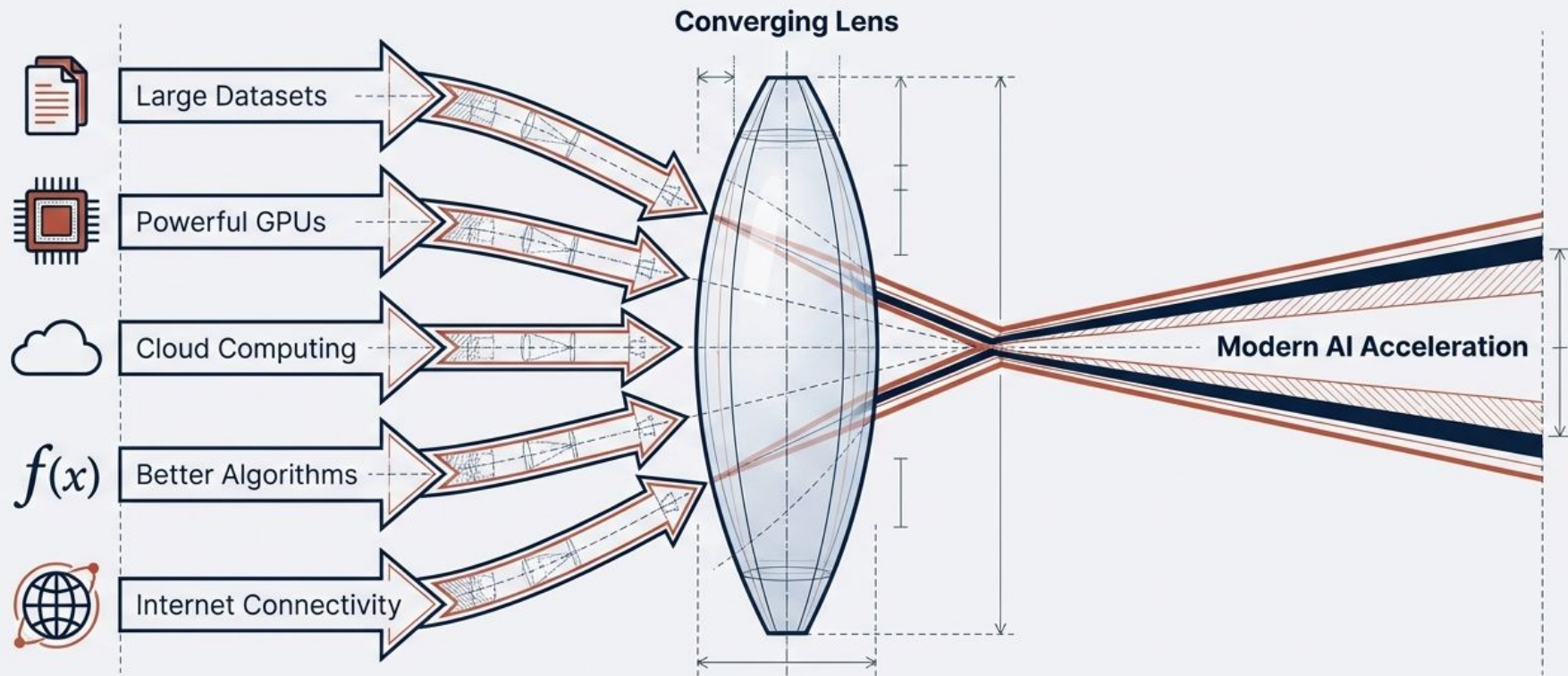
Generative AI

The capability moves from analyzing existing data to creating new content. Examples: ChatGPT, DALL·E.



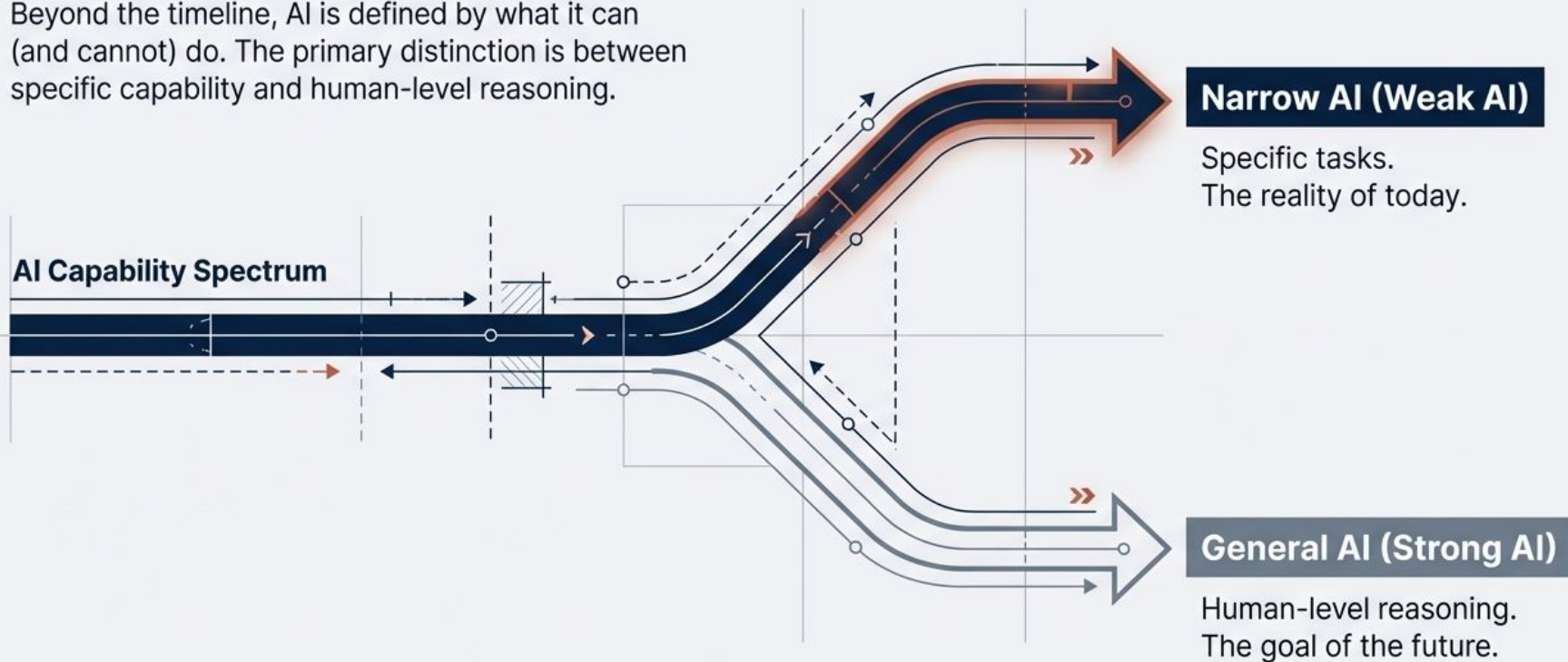
Why Progress Accelerated Recently

Modern AI was made possible by the convergence of five specific drivers.



Classifying AI by Capability

Beyond the timeline, AI is defined by what it can (and cannot) do. The primary distinction is between specific capability and human-level reasoning.



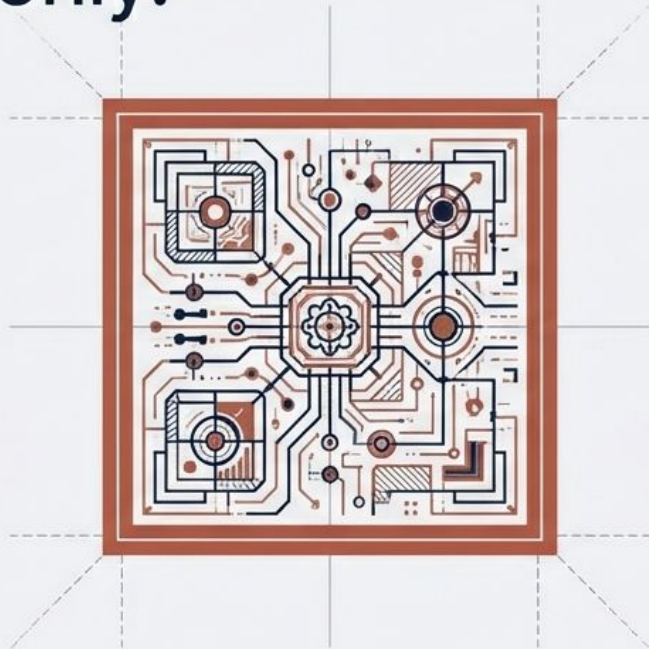
Note: 'Super AI' remains a theoretical future concept beyond General AI.

Narrow AI (Weak AI)

Artificial Intelligence designed for specific tasks only.

The Constraint

Key Limitation: It cannot think beyond its specific training data or programming. It simulates intelligence within a bounded context but possesses no consciousness or genuine reasoning.



The Status

Status: Today's AI is almost entirely Narrow AI.

Core Characteristics of Narrow AI

- Task-Specific Intelligence
- Learns From Data (Not From Experience Like Humans)
- No Transfer of Intelligence
- Works Inside Fixed Boundaries

Narrow AI in Practice

Even the most impressive modern tools fall under this category.



Utility

Google Maps (Routing),
Face Recognition.



Entertainment

Netflix Recommendations
(Pattern matching).



Assistants

Alexa, Siri
(Voice processing).



Generative

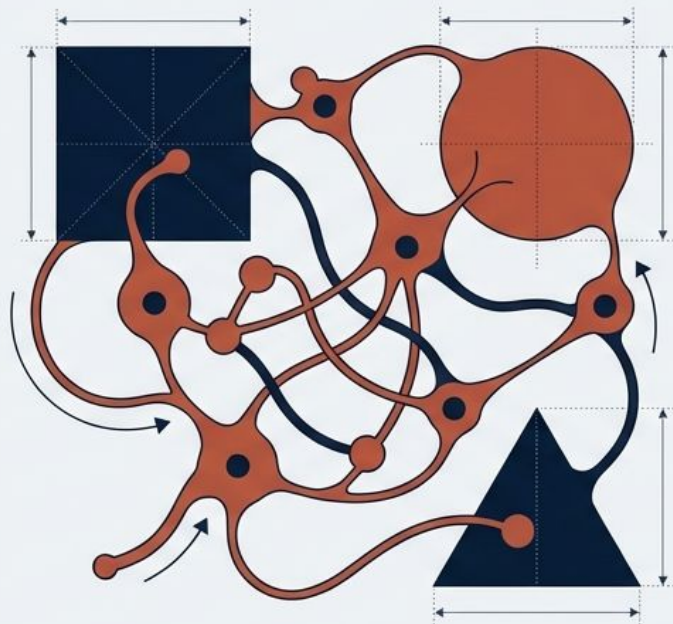
ChatGPT (Text prediction).
**Note: Despite its power,
this is still Narrow AI.*

General AI (Strong AI)

Machine intelligence comparable to a human brain.

Key Capabilities

- ✓ Can learn any task a human can.
- ✓ Reasons like humans.
- ✓ Transfers knowledge from one domain to another without retraining to another without retraining.



Cross-Domain Knowledge Transfer: Seamlessly applying learning from one context to another.

Core Characteristics of General AI

- Human-Level Flexibility
- Transfer Learning Across Domains
- Autonomous Learning
- True Reasoning + Understanding
- Common Sense

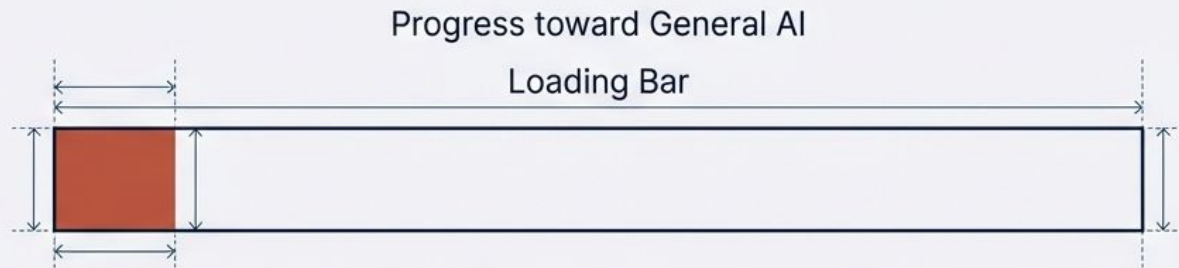
The Current State of General AI

Status

**None
Exist Yet**

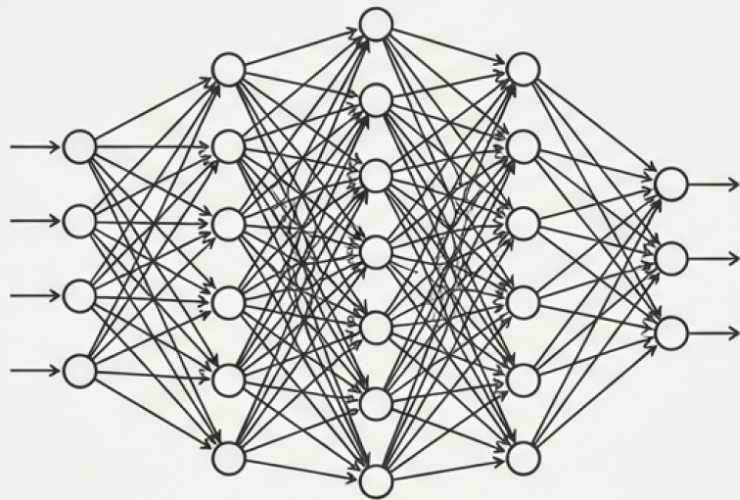
Context

While Narrow AI has mastered specific domains (chess, writing, protein folding), General AI remains a research goal.



Data-Centric Modeling: The ROI of Quality

Why the next leap in AI performance comes from your dataset, not your algorithm.



Traditional View: Architecture Focused



Modern View: Data Focused

Core Thesis: Better Data = Better AI.

Data-Centric Modeling

The Plateau of Model-Centric AI

Definition: The Model-Centric approach involves holding the training data fixed while iteratively improving the code architecture.

Common Tactics:

- Changing model families (e.g., ResNet to EfficientNet)
- Extensive hyperparameter tuning
- Increasing model complexity (adding layers/parameters)

The Problem:

When the underlying data is **noisy**, even the most advanced architectures fail to generalize. We are currently **over-optimizing** for **noise** rather than **signal**.

The Golden Rule of Modern ML

Simple Model + Clean Data > + Complex Model + Bad Data

The Enemies of Performance

Incorrect Labels

The model learns falsehoods from the start.

Duplicate Samples

Causes overfitting to specific, repeated examples.

Noisy Inputs

Obscures the signal, confusing the algorithm.

Missing Values

Introduces ambiguity in the decision boundary.

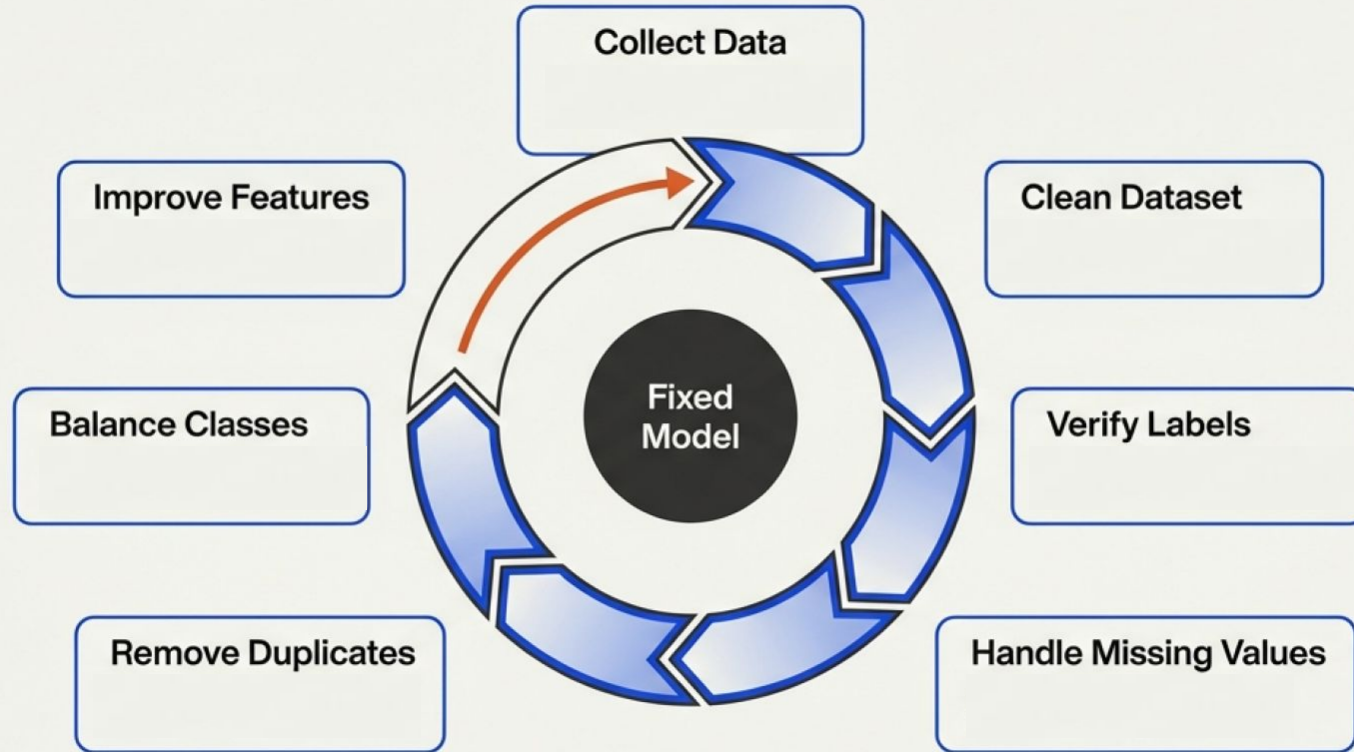
Imbalanced Classes

Biases predictions toward the majority class.

Result:

Even Deep ResNets or Transformers fail when subject to these flaws.

The Data-Centric Workflow



Unlike linear software development, AI is a continuous cycle of data refinement.